

FISHER BRANCH, MB, Canada

WMO: 714420

Lat: 51.0834N

Lon: 97.5544W

Elev: 830

StdP: 14.26

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-28.3	-23.1	-41.1	0.5	-24.4	-37.1	0.7	-22.5	21.5	7.6	19.9	7.4	4.4	290	0.648

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	21.6	84.7	68.5	81.6	66.9	78.9	65.0	71.6	80.6	69.2	78.1	67.3	75.9	9.5	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
68.4	107.5	77.0	66.0	98.8	74.4	63.9	91.6	72.2	35.9	80.5	33.9	78.1	32.2	76.1	80.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 20.1	17.9	16.0	DB	-35.9	90.5	5.4	3.8	-39.8	93.2	-43.0	95.4	-46.0	97.6	-49.9	100.3	(4)
(5)			WB	-35.3	75.6	4.5	2.7	-38.6	77.5	-41.2	79.1	-43.7	80.6	-47.0	82.5	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	35.6	0.4	4.9	19.4	36.8	49.6	60.3	65.4	63.5	54.2	40.3	23.1	7.2	(6)
(7)		DBStd	25.28	14.74	13.81	14.20	11.48	8.99	7.08	5.85	6.19	8.06	8.65	12.15	14.22	(7)
(8)		HDD50	6797	1538	1263	949	413	120	8	0	0	47	323	808	1328	(8)
(9)		HDD65	10926	2003	1683	1413	847	483	173	67	104	335	767	1258	1793	(9)
(10)		CDD50	1530	0	0	1	16	107	316	479	418	172	21	0	0	(10)
(11)		CDD65	184	0	0	0	0	5	31	81	57	10	0	0	0	(11)
(12)		CDH74	2015	0	0	0	4	110	348	790	629	128	6	0	0	(12)
(13)		CDH80	436	0	0	0	0	23	64	173	151	23	2	0	0	(13)
(14)	Wind	WSAvg	7.4	7.2	7.2	7.6	8.1	8.2	7.3	6.4	6.6	7.3	8.1	7.7	7.2	(14)
(15)	Precipitation	PrecAvg	21.8	1.2	0.7	1.3	1.3	2.5	3.3	2.6	2.5	2.2	1.7	1.3	1.4	(15)
(16)		PrecMax	26.2	2.5	1.7	4.4	2.9	5.2	5.7	5.6	5.0	4.0	3.2	3.8	2.6	(16)
(17)		PrecMin	16.9	0.2	0.2	0.4	0.3	0.9	1.7	1.2	1.3	0.8	0.1	0.2	0.4	(17)
(18)		PrecStd	2.8	0.7	0.4	0.9	0.7	1.3	1.2	1.0	0.9	0.9	0.8	0.9	0.5	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	37.8	38.5	56.6	73.2	83.1	87.7	88.1	89.3	83.4	70.9	55.0	38.5	(19)
(20)			MCWB	33.9	33.9	47.6	54.4	60.8	70.6	71.2	71.0	66.0	57.2	46.3	33.7	(20)
(21)		2%	DB	32.3	34.6	46.8	66.1	76.6	81.4	84.4	83.9	77.6	64.1	47.6	33.5	(21)
(22)			MCWB	30.3	32.4	40.2	50.1	57.4	65.6	69.8	68.4	63.1	52.7	41.4	30.6	(22)
(23)		5%	DB	27.0	31.7	41.3	60.7	71.6	78.2	81.5	80.6	73.2	58.8	42.8	30.0	(23)
(24)			MCWB	25.1	29.5	36.6	47.8	55.4	63.8	67.9	66.7	60.6	50.2	38.0	28.2	(24)
(25)		10%	DB	21.9	26.0	37.2	55.0	66.6	74.8	78.6	77.3	68.8	54.1	38.4	26.7	(25)
(26)			MCWB	20.4	23.9	33.7	44.5	52.7	61.9	66.1	64.9	58.8	48.2	35.1	25.2	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	34.2	34.7	46.9	55.8	64.2	73.8	75.6	73.4	69.0	58.9	47.8	34.0	(27)
(28)			MCD B	36.8	37.1	53.2	69.5	76.9	82.8	84.2	84.2	77.9	68.3	53.3	37.6	(28)
(29)		2%	WB	30.5	32.5	40.3	51.4	60.3	68.8	72.2	70.7	65.4	54.9	42.2	31.1	(29)
(30)			MCD B	32.6	34.6	45.9	64.8	70.9	77.5	80.5	80.4	73.4	61.0	46.7	33.3	(30)
(31)		5%	WB	25.3	29.2	36.8	48.3	57.1	66.0	69.8	68.5	62.7	51.3	38.6	28.3	(31)
(32)			MCD B	27.1	31.6	40.9	59.0	67.7	74.7	78.4	77.9	70.2	57.9	42.5	29.9	(32)
(33)		10%	WB	20.5	24.0	34.0	45.3	54.5	63.6	67.7	66.6	60.0	47.9	35.1	25.1	(33)
(34)			MCD B	21.9	25.9	36.9	54.8	65.0	71.7	76.1	74.9	67.5	53.8	38.3	26.7	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	17.3	19.3	19.3	21.3	22.8	21.0	21.6	22.3	20.9	16.4	14.2	15.6	(35)
(36)			MCDBR	17.9	16.8	19.1	29.8	30.6	26.0	24.9	26.9	27.9	25.0	20.2	17.1	(36)
(37)			MCWBR	16.2	14.8	13.8	17.1	15.8	13.4	13.1	13.6	15.3	15.3	14.4	15.3	(37)
(38)		5% WB	MCDBR	17.8	16.4	17.7	27.7	24.8	21.9	22.3	23.5	23.1	22.1	18.6	16.2	(38)
(39)			MCWBR	16.3	14.6	13.0	16.4	14.3	12.6	13.1	13.0	14.2	14.3	13.5	14.6	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.259	0.283	0.307	0.354	0.364	0.380	0.391	0.373	0.342	0.297	0.264	0.242	(40)	
(41)		taud	2.244	2.193	2.260	2.324	2.371	2.356	2.348	2.405	2.479	2.559	2.448	2.312	(41)	
(42)		Ebn at Noon	243	267	283	281	282	278	272	272	270	266	246	235	(42)	
(43)		Edh at Noon	21	30	34	36	36	37	37	33	28	21	18	17	(43)	
(44)	All-Sky Solar Radiation	RadAvg	356	713	1132	1587	1738	1833	1896	1577	1111	637	338	262	(44)	
(45)		RadStd	33	48	79	106	144	143	100	111	108	78	46	25	(45)	

Historical Trends

		Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	N/A	N/A	+5.42	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (19 neighbors)	N/A	N/A	N/A	-1.04	-1.44	N/A	N/A	N/A	N/A

Nomenclature: See separate page